

Kevin Niu

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Engineering

Atlanta, GA

May 2024 – May 2027 (Expected)

EXPERIENCE

Combat Robotics Finance Chair, 12lb Chassis Lead, 3lb Team Captain

May 2024 – Present

RoboJackets, GT

Atlanta, GA

- Designed a hub-motor drum spinner system using an integrated sensorless brushless powertrain, achieving a 12% weight reduction and 15% higher energy storage compared with previous belt-driven designs
- Led an interdisciplinary team of 10+ members to design and manufacture a 12lb combat robot while teaching CAD, soldering, and manufacturing fundamentals to less experienced members
- Captained a rookie 3lb combat robot to place 3rd among 21 robots at a local tournament and competed nationally
- Integrated brushless motors, electronic speed controllers, batteries, receivers, and mechanical assemblies into competition-ready combat robots and diagnosed electrical and drivetrain failures during testing
- Managed a \$9,000 budget for 100+ members across 9 teams, securing and allocating funding from corporate sponsors and the Student Organization Finance Office while managing reimbursements and purchase requests
- Oversaw procurement for all combat robotics teams and coordinated with Georgia Tech accountants to purchase and deliver electronics and raw materials to an off-campus workshop

Circuit Design Team

Aug 2023 – May 2024

Illinois Electric Motorsports, UIUC

Champaign, IL

- Designed a 4-layer vehicle sensor interface PCB in Altium using an STM32 microcontroller and CAN protocol for distributed vehicle telemetry, implementing low-pass filters and op-amps for signal conditioning of analog sensors
- Led a 3-member validation sub-team developing an Arduino-based current-RPM test rig using a Hall-effect current sensor, achieving a 20% current draw reduction through fan-curve optimization
- Led assembly of high-voltage accumulator modules using pouch cells, thermistors, and voltage-sense boards
- Validated accumulator systems using an oscilloscope, logic analyzer, and CAN telemetry logs to confirm sensor functionality and zero observed packet loss during testing

Combat Robotics Electrical Lead, RoboBrawl Committee

Aug 2023 – May 2024

iRobotics, UIUC

Champaign, IL

- Developed an embedded wireless system using ESP32 and ESP-NOW to display wins, losses, and ready status for judge-competitor communication, reducing match downtime by 20%
- Implemented an asynchronous web server using AJAX to provide real-time competition status updates
- Directed a team of 15+ members to source and integrate electronics for a 30lb combat robot while tuning firmware parameters to enhance drivetrain consistency, reducing motor failure rate and improving startup commutation

PROJECTS

OpenTurner

July 2025

- Built a battery-powered ESP32-C3 BLE HID wireless page-turner with ADC-based battery monitoring, sleep-mode power management, and dynamic device naming
- Developed C++ firmware using non-blocking event handling, hardware debouncing, and persistent NVS configuration for responsive input processing and on-board memory of device settings

Analog PID Car

May 2024

- Designed analog PID control circuitry using operational amplifiers for proportional, integral, and derivative feedback for autonomous braking and distance control
- Tuned PID constants through oscilloscope analysis to achieve less than 2% steady-state error and zero overshoot

TECHNICAL SKILLS

Programming: C++, C, Java, Python, MATLAB, JavaScript, Node.js, SQL

Developer Tools: Git, Docker, VS Code, PyCharm, IntelliJ, Eclipse

Embedded Systems: STM32, ESP32, Arduino, Raspberry Pi, FreeRTOS, CAN bus, I²C, SPI, UART, PWM

Hardware: PCB Design, Altium Designer, VHDL, Verilog, FPGA, Altera Quartus, Oscilloscope, Logic Analyzer,

Mechanical: SOLIDWORKS (CSWP), Fusion 360, Inventor, 3D Printing, 2-Axis Mill, Lathe, CNC Router, Waterjet

Hobbies: Pickleball, Poker, Photography, Skating, Woodworking